Tetromino tilings with line clears

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Abstract

Walkup proved in 1965 that a $w \times n$ rectangle can be tiled by T-tetrominoes exactly when $4 \mid w$ and $4 \mid n$, and a line of later work measured how badly other rectangles fail. We ask the same question in a *dynamic* setting: pieces fall one at a time under gravity, and any row that becomes full is deleted. A *perfect clear* is a play that starts from the empty board and returns to it.

Line clears collapse the static obstruction almost completely. Four of the seven tetromino types — I, T, J and L — admit a perfect clear at every width $w \geq 4$, and O admits one at every even width; we determine the minimum number of pieces in each case, for all widths. The two exceptions are S and Z, which admit no perfect clear at any width, alone or together. The classification extends to every subset of the seven types.

The contrast with the static theory is sharpest for T, where Walkup's condition reappears *inside* the dynamic construction: the part of the construction that uses no line clears is exactly a static tiling, so its width must be a multiple of four, and the line clears cover precisely the residues that static tiling forbids.

1 Introduction

A tetromino is a polyomino with four cells; there are seven of them up to translation, conventionally named I, O, T, S, Z, J and L. Which rectangles can be tiled by copies of a single one?

For the T-tetromino the answer is a classical theorem of Walkup [1]: a $w \times n$ rectangle admits a tiling by T-tetrominoes if and only if $4 \mid w$ and $4 \mid n$. Korn and Pak [3] later analysed the structure of those tilings, and Reid [5] placed questions of this kind in the general setting of Klarner systems. Rectangles that fail Walkup's condition cannot be tiled at all, so Hochberg [2] asked how close one can get: the *gap number* is the least number of monominoes that must be left uncovered. He proved lower bounds of a strikingly rigid kind — width 5 forces at least one monomino every five columns, width 7 one every seven columns — and showed the gap number of T is one of $5, 6, 7, 9$; Torres and Vallejo [4] later brought the cap down to six.

All of that is static. The tetrominoes are also the pieces of Tetris, and there the board is not a passive region: pieces fall under gravity, and a row that becomes completely full is deleted, with everything above it dropping by one. The natural question is what those deletions buy.

Definition 1. A *perfect clear* at width w is a play that begins with an empty board of width w and, after finitely many placements and the deletions they trigger, ends with the board empty again.

Our main result is that line clears dissolve the static obstruction for five of the seven pieces and for no others.

piece	static	dynamic (this paper)
I	$4 \mid w$, $4 \mid n$ needed for a full tiling	every width, $n = w/4$ or w
O	only even widths	every even width, $n = w/2$
T	Walkup: $4 \mid w$ and $4 \mid n$	every width $w \geq 4$, $n = w$
J, L		every width $w \geq 4$, $n = w/2$, w or $2w$
S, Z	gap number conjectured unbounded [2]	**never, at any width**

The T row is where the two theories meet most sharply, and it is worth stating the mechanism now, because it is the shape of the whole paper. Our construction for T (Theorem 7) splits the board into column-disjoint *units*. All but the last unit must fill their rows without triggering any clear — that is, they must *statically tile* a $x \times 4$ rectangle, so Walkup forces $4 \mid x$. Widths 5, 6 and 7 therefore cannot be built this way at all. They are exactly the widths the final unit handles, and the final unit is the one allowed to use line clears. The dynamic rule covers precisely the residues the static theorem forbids.

Two further remarks place this work. Hoogeboom and Kosters [8] characterised which board configurations are reachable from empty when *all seven* piece types are available; with the full set the only obstruction is a divisibility condition on the number of cells, so the question is interesting only when the supply of pieces is restricted. That is the case we treat. Separately, Tetris is hard: deciding how much of a given board can be cleared is NP-complete [9], and this remains true for a single piece type [10]; Baccherini and Merlini [11] study the combinatorics of related games. Our question is different in kind — not the complexity of clearing a given board, but which piece sets can empty an *empty* board at all — and it turns out to have a clean answer.

2 The model

We fix the rules precisely, because one of our results is that they matter less than one might expect.

The board of width w is a finite set $B \subseteq \{0, \ldots, w-1\} \times \mathbb{Z}_{\geq 0}$ of occupied cells; we write (r, c) for the cell in row r and column c, with row 0 at the bottom. Play starts from $B = \emptyset$. A move chooses a piece type from the available set P, chooses one of its rotations and a horizontal position, and places its four cells as a set X, subject to:

(L1) $X \cap B = \emptyset$ — the four cells were empty;

(L2) X shifted down by one row is not a legal position — the piece cannot fall further;

(L3) after the placement, every full row is deleted simultaneously, and each surviving cell descends by the number of deleted rows below it.

Condition (L2) says a piece locks only when supported. It says nothing about how the piece reached its position. Real implementations differ here: hard drop (choose a column, fall straight down), soft drop with horizontal movement (allowing tucks), and rotation systems with wall kicks (allowing spins) permit successively more placements. Writing HD $\subseteq$ SD $\subseteq$ SRS for the three sets of reachable placements, our results are insensitive to the choice:

- every construction in this paper was found under hard drop only, so it remains valid in the larger models;
- every impossibility proof uses only (L1), (L2), (L3) and the shape of the pieces, never the path by which a piece arrived, so it holds in all three.

Finally, if a perfect clear uses n pieces and deletes R rows in total, then counting cells gives

$$4n = Rw, \tag{1}$$

since every cell that is ever placed is eventually deleted, and each deletion removes w cells.

3 Two tools

The first tool does almost all of the work.

Lemma 2 (Column counting)**.** *In a perfect clear at width w that deletes R rows in total, every column receives exactly R cells.*

Proof. A cell never changes column: placement fixes its column, and the descent in (L3) is vertical. A deletion removes one cell from each of the w columns, since the deleted row is full. The board ends empty, so every cell placed in column c is eventually deleted, and deletions remove cells from column c one at a time, once per deleted row. Hence column c receives exactly R cells. □

Lemma 2 is purely combinatorial: it uses (L3) and nothing else, so it is independent of the rotation system. Summing over columns recovers (1).

To use it we record, for each piece type, its *column profiles*: the tuple of cells contributed to consecutive columns, one tuple per rotation.

piece	column profiles
I	$(1,1,1,1)$, (4)
O	$(2,2)$
T	$(1,2,1)$, $(1,3)$, $(3,1)$
S, Z	$(1,2,1)$, $(2,2)$
J, L	$(2,1,1)$, $(1,1,2)$, $(1,3)$, $(3,1)$

If x_c denotes the number of pieces placed with a given profile starting at column c, Lemma 2 becomes a system of linear equations in the x_c, one per column, each with right-hand side R. We refer to it as the *counting system*. Note that S and Z have the same profiles, as do J and L; counting alone can never distinguish them. The reason is structural.

Lemma 3 (Mirror)**.** *The board and the rules are symmetric under left–right reflection, which exchanges J with L and S with Z. Consequently J and L behave identically at every width and every piece count, as do S and Z.*

Proof. Reflecting every placement of a legal play gives a legal play: (L1) and (L2) are preserved because reflection is a bijection on cells respecting rows, and a row is full if and only if its reflection is. Reflection maps the rotations of J onto those of L and the rotations of S onto those of Z. □

4 Single piece types

Throughout this section P consists of one type.

4.1 S and Z

Theorem 4. *S alone admits no perfect clear, at any width. The same holds for Z alone.*

Proof. Let h_c be the number of horizontal S pieces (profile $(1,2,1)$) whose leftmost column is c, and v_c the number of vertical ones (profile $(2,2)$). The counting system reads

$$h_c + 2h_{c-1} + h_{c-2} + 2v_c + 2v_{c-1} = R \qquad (0 \le c \le w-1),$$

with the convention that a variable vanishes when its piece would leave the board. Column 0 gives $h_0 + 2v_0 = R$ and column 1 gives $h_1 + 2h_0 + 2v_1 + 2v_0 = R$; subtracting, $h_1 + h_0 + 2v_1 = 0$, so $h_0 = h_1 = v_1 = 0$ and $2v_0 = R$. The same subtraction at columns $c+2, c+3$ propagates this: by induction $h_c = 0$ for all c, $v_c = 0$ for odd c, and $v_c = R/2$ for even c.

If w is odd, then $w-1$ is even, so $v_{w-1} = R/2$; but a vertical piece starting at column $w-1$ would occupy column w, so $v_{w-1} = 0$ and $R = 0$. There is no nonempty play.

If w is even the counting system is consistent, and we need the geometry. Every piece is a vertical S starting at an even column, so the columns split into independent blocks $\{2k, 2k+1\}$, no piece straddling two blocks. A vertical S occupies relative rows $\{1,2\}$ in its left column and $\{0,1\}$ in its right one: only the *right* column reaches relative row 0. Since every piece starts at an even column, every cell in an even column sits at relative row at least 1 above the piece's base, and the base is at absolute row ≥ 0; so no placement ever fills cell $(0,0)$.

Nor can a cell descend into it. A cell reaches row 0 only if all the rows below its current position are deleted, which requires row 0 itself to be deleted, which requires $(0,0)$ to be occupied. The two statements — "$(0,0)$ is never filled" and "row 0 is never deleted" — therefore support each other, and a simultaneous induction on the moves establishes both.

The first piece placed in the block $\{0,1\}$ rests on the floor, since nothing else can occupy those columns, and so leaves a cell in row 0. Row 0 is never deleted, and a cell leaves the board only when its row is deleted; that cell is permanent, and the board never empties. The statement for Z follows from Lemma 3. □

4.2 O and I

Theorem 5. *O alone admits a perfect clear exactly at even widths, with $n = w/2$ pieces and $R = 2$.*

Proof. With o_c the number of O pieces starting at column c, the counting system is $2o_c + 2o_{c-1} = R$. Column 0 gives $2o_0 = R$, column 1 gives $2o_1 = 0$, and inductively $o_c = R/2$ for even c and 0 for odd c. If w is odd then $o_{w-1} = 0$ because the piece would leave the board and $o_{w-2} = 0$ because $w-2$ is odd, so column $w-1$ gives $R = 0$. If w is even, place $w/2$ pieces side by side: rows 0 and 1 fill and are deleted. That $R = 2$ cannot be improved is part of Theorem 10. □

Theorem 6. *I alone admits a perfect clear at every width w, with $n = w/4$ pieces and $R = 1$ when $4 \mid w$, and $n = w$ pieces and $R = 4$ otherwise.*

Proof. For the construction, place one vertical I in each column: rows 0 to 3 fill and are deleted, using $n = w$ pieces. If $4 \mid w$, lay $w/4$ horizontal I pieces along row 0 instead, deleting one row. The minimality of R in each case follows from Theorem 10 below. □

4.3 T

This is the case where the static theory reappears, so we give it in full.

Theorem 7. *T alone admits a perfect clear at every width $w \ge 4$, and the minimum number of pieces is exactly $n = w$, with $R = 4$.*

Proof of the lower bound. By (1) it suffices to show $R \geq 4$. Let h_c, p_c, q_c count the pieces with profile $(1,2,1)$, $(1,3)$ and $(3,1)$ respectively starting at column c. The counting system is

$$N_j := h_j + 2h_{j-1} + h_{j-2} + p_j + 3p_{j-1} + 3q_j + q_{j-1} = R.$$

We use only $j = 0, 1, 2$, which exist as soon as $w \geq 3$.

If $R = 1$, then $N_0 = 1$ forces $(h_0, p_0, q_0) = (1,0,0)$ or $(0,1,0)$; the first gives $N_1 \geq 2h_0 = 2$ and the second $N_1 \geq 3p_0 = 3$, both exceeding 1.

If $R = 2$, then $3q_0 \leq 2$ forces $q_0 = 0$ and $(h_0, p_0) \in \{(2,0),(1,1),(0,2)\}$; then $N_1 \geq 2h_0 + 3p_0$ equals 4, 5 or 6, all exceeding 2.

If $R = 3$, the solutions of $N_0 = 3$ are $(3,0,0)$, $(2,1,0)$, $(1,2,0)$, $(0,3,0)$ and $(0,0,1)$. For the first four, $N_1 \geq 2h_0 + 3p_0 \geq 6$. The remaining case has $q_0 = 1$ and $h_0 = p_0 = 0$, whence $N_1 = h_1 + p_1 + 3q_1 + 1 = 3$, so $q_1 = 0$ and $(h_1, p_1) \in \{(2,0),(1,1),(0,2)\}$; but then $N_2 \geq 2h_1 + 3p_1 \geq 4 > 3$.

So $R \geq 4$ and $n = Rw/4 \geq w$. □

Proof of the construction. Write the four rotations of T as

$$\mathsf{U} = \{(0,0),(0,1),(0,2),(1,1)\}, \quad \mathsf{D} = \{(0,1),(1,0),(1,1),(1,2)\},$$

$$\mathsf{L} = \{(0,1),(1,0),(1,1),(2,1)\}, \quad \mathsf{R} = \{(0,0),(1,0),(1,1),(2,0)\},$$

with cells given as (row, column) offsets from the landing position, and write $\mathsf{X}@c$ for a placement of X whose leftmost column is c.

We decompose the board into units occupying disjoint intervals of columns. Because a piece never leaves its unit's columns, the landing row of each placement, and whether a row of that unit is full, depend only on that unit's own contents. Two kinds of unit are needed, with $R = 4$ throughout:

The stacking unit S, of width 4, is U@0, L@2, R@0, D@1. With clears switched off it fills rows 0 to 3 of its four columns exactly, using 4 pieces.

The final units E_m, of width $m \in \{4,5,6,7\}$ and m pieces:

E_4 : U@0 L@2 R@0 D@1
E_5 : U@0 D@2 R@0 L@3 D@1
E_6 : U@1 L@4 D@2 R@0 D@0 D@3
E_7 : U@0 U@3 D@2 L@5 D@4 R@0 D@1

With clears switched on, each empties its own columns after deleting four rows.

Now write $w = 4a + m$ with $m \in \{4,5,6,7\}$ and $a \geq 0$; this is possible for every $w \geq 4$, taking $m \equiv w \pmod 4$ in the stated range. Play a copies of S in the columns $[0, 4a)$, then E_m in the columns $[4a, w)$.

While the copies of S are played, the columns of E_m are empty, so no row of the board is full and no deletion occurs; each copy of S therefore fills rows 0 to 3 of its own columns, as required. When E_m begins, every column outside it is full in rows 0 to 3, so a row of the board is full exactly when E_m fills that row within its own columns. Hence E_m evolves exactly as it would on a board of width m on its own, deleting four rows; those deletions remove the rows of the copies of S as well, and the board is empty. The piece count is $4a + m = w$, matching the lower bound. □

Remark 8 (Where Walkup reappears). The requirement on a stacking unit — fill rows 0 to 3 of x columns with no deletion — says exactly that the $x \times 4$ rectangle admits a static tiling by T-tetrominoes. By Walkup's theorem this forces $4 \mid x$. So widths 5, 6 and 7 admit no stacking unit at all, and those are precisely the widths carried by the final unit, the one permitted to use line clears. The dynamic rule covers exactly the residues the static theorem forbids.

Walkup's theorem does not, however, settle whether the 4×4 tiling can be *reached* under gravity: support is not a concern of static tiling, and a tiling in which some piece must be inserted beneath another would be unusable here. That it can be reached — by the four placements listed above — is the one genuinely dynamic ingredient of the construction.

4.4 J and L

Theorem 9. *J alone admits a perfect clear at every width $w \geq 4$, with minimum piece count*

$$n = \begin{cases} w/2 & (w \equiv 0 \bmod 4), \quad R = 2, \\ w & (w \equiv 2 \bmod 4), \quad R = 4, \\ 2w & (w \text{ odd}), \quad R = 8. \end{cases}$$

The same holds for L.

Sketch. The construction is again a unit decomposition, with $\mathsf{A} = (2,1,1)$, $\mathsf{B} = (1,1,2)$, $\mathsf{P} = (1,3)$ and $\mathsf{Q} = (3,1)$ denoting the rotations of J by their profiles. The units are

AB	width 4,	$R = 2$,	A@0, B@1
PQ	width 2,	$R = 4$,	P@0, Q@0
PQPQ	width 2,	$R = 8$,	(P@0, Q@0) twice
U_5	width 5,	$R = 8$,	A@0 P@1 P@0 P@3 Q@0 Q@3 A@2 A@0 P@3 B@1

and the board is built as $\mathsf{AB}^{w/4}$ when $4 \mid w$, as $\mathsf{PQ}^{w/2}$ when $w \equiv 2 \pmod 4$, and as U_5 followed by $(w-5)/2$ copies of PQPQ when w is odd. The verification is the same two steps as in Theorem 7 and is a finite check on four units, independent of w. Minimality follows from Theorem 10. L follows by Lemma 3. □

5 Lower bounds at every width

The minimality claims above assert, for each piece and each width, that the counting system has no solution for smaller R. For a fixed width this is a finite check. The following observation upgrades it to all widths at once.

Theorem 10. *For each piece type and each R, the set of widths at which the counting system is solvable is eventually periodic in w, and the period can be computed. The resulting minimum values of R are*

piece	*minimum R*
I	1 *if* $4 \mid w$, *else* 4
O	2 *if w even; no solution if w odd*
T	4 *at every width*
S, Z	2 *if w even; no solution if w odd*
J, L	2 *if* $4 \mid w$; 4 *if* $w \equiv 2 \bmod 4$; 8 *if w odd*

Proof. Sweep the columns from left to right, maintaining as state the vector of contributions already committed to the columns not yet processed. A profile spans at most four columns, so the state has at most three components, each an integer in $[0, R]$ — any larger value can never be reduced and is dead. The state space is therefore finite. Processing one column is a fixed relation on states, so it induces a deterministic map on *sets* of states; iterating it from the initial set $\{(0, \ldots, 0)\}$ gives a sequence of subsets of a finite set, which must eventually cycle.

The rightmost few columns are special — a profile may no longer fit — but the sequence of restricted maps applied there depends only on how many columns remain, not on w. Hence

solvability at width w is obtained by applying a fixed tail map to the $(w-k)$-th term of an eventually periodic sequence, where $k \leq 3$ is the number of special columns, and is therefore eventually periodic in w. Computing the pre-period and period gives the table. □

The entries for S and Z deserve emphasis, because they explain the shape of Theorem 4.

Corollary 11. *At every even width, S alone passes the counting test with $R = 2$. Consequently no argument based on Lemma 2 alone can rule out S, and the geometric step in the proof of Theorem 4 is unavoidable.*

Among all seven pieces, this is the only place where counting and geometry part company: for I, O, T, J and L the counting bound is achieved by an explicit construction, and for S and Z at odd widths counting already gives a contradiction. The single gap is S and Z at even widths, and it is exactly the gap Theorem 4 fills.

6 Sets of pieces

Suppose now that a set P of types is available, and the player may use any of them in any order.

Theorem 12. *$\{S, Z\}$ together admits no perfect clear, at any width.*

Sketch. S and Z have the same column profiles, so the counting argument of Theorem 4 applies verbatim to their total counts: no horizontal piece is used, every piece is vertical and starts at an even column, and odd widths are excluded outright. At even widths the columns again split into independent blocks $\{2k, 2k+1\}$, and the width-w problem reduces to a width-independent question about one block.

Let F_0, F_1 be the sets of filled rows in the two columns of a block and put $d_j = |F_0 \cap [0, j]| - |F_1 \cap [0, j]|$. Placing a vertical Z at height r increments d_r and d_{r+1}; placing a vertical S decrements them; deleting row m removes d_m from the sequence and changes no other value. One then checks four invariants of the reachable states: $|F_0| = |F_1|$; the occupied rows form an initial segment; $|d_j| \leq 1$ for all j; and $d_0 \neq 0$ whenever the block is nonempty. The last one says row 0 of the block is never full, so it is never deleted, and the cell the first piece leaves in row 0 is permanent. □

Theorem 13. *Let $w \geq 4$. A set P of tetromino types admits a perfect clear at width w if and only if P meets $\{I, T, J, L\}$, or w is even and $O \in P$.*

Proof. If P contains one of I, T, J, L, use that type alone (Theorems 6, 7, 9); each of those constructions is available at every $w \geq 4$. Otherwise $P \subseteq \{O, S, Z\}$, and the profiles of all three lie in $\{(1, 2, 1), (2, 2)\}$, so the counting derivation in the proof of Theorem 4 applies to their total counts without change: horizontal pieces are unused, every piece starts at an even column, and at odd widths $R = 0$. At even widths, if $O \in P$ then $w/2$ copies of O clear two rows; if not, then $P \subseteq \{S, Z\}$ and Theorems 4 and 12 apply. □

Remark 14 (Widths below four). The hypothesis $w \geq 4$ is needed, and T is the reason. I clears every width from 1 upwards and J and L clear widths 2 and 3, so those constructions lose nothing; but T alone clears neither width 2 nor width 3, although the counting system permits $R = 4$ at both. (Width 1 admits no T placement at all, since every rotation of T spans at least two columns.) The classification at $w \leq 3$ would therefore need T removed from the list, and we have not determined which mixed sets containing T succeed there.

Remark 15. It is tempting to argue that since $\{S, Z\}$ is the only unusable pair, every set of three or more types must work. That is false: $\{O, S, Z\}$ has three types and fails at every odd width. The reason is that O carries a width restriction of its own, so "only one bad pair" says nothing about sets built from the three restricted types.

7 Restricted supply: the seven-bag randomiser

Modern Tetris does not let the player choose the next piece. Pieces arrive in *bags*: each block of seven consecutive pieces is a permutation of the seven types. A single slot of *hold* lets the player set one arriving piece aside.

This changes the question, because every construction above repeats one type.

Theorem 16. *Under the seven-bag constraint, with the order inside each bag at the player's choice, the minimum number of pieces in a perfect clear is*

w	4	5	6	7	8	9	10	11	12	13	14
unrestricted	1	5	3	7	2	9	5	11	3	13	7
seven-bag	1	5	3	7	4	9	10	11	9	13	14

The pattern is that the constraint is free at odd widths and costly at even ones. At an odd width the unrestricted minimum already has $R = 4$, and four rows give room to place seven different types; at an even width the unrestricted optimum is one or two rows, and a thin region cannot absorb variety. A single row admits only horizontal I pieces, so $w/4 \geq 2$ is immediately fatal; two rows survive to $w = 8$ and fail from $w = 10$.

Width 16 continues the pattern and is the most expensive case we have settled: the unrestricted minimum is 4, and the seven-bag minimum is 12. Both $n = 4$ and $n = 8$ are impossible — every admissible multiset was decided — while $n = 12$ admits five, among them one copy each of I and L with two of every other type.

The harder question is what the *order* inside the bags costs, and here we can report less. Fix $w = 10$ and $n = 10$, the setting of the standard opener. Ten placements consume at most eleven arrivals, so only the first four pieces of the second bag matter, giving $P(7, 4) = 840$ prefixes to consider for each of the $7! = 5040$ orders of the first bag.

A sample of 100 first bags, all 840 prefixes each, gave nine failures in three bags — a rate of 1.1×10^{-4} over the sample — and a fourth bag, TJLISZO, had been settled earlier with four. Those thirteen were all we had when the structural results below were found; the exhaustive sweep has since superseded them, and we quote it instead.

Two structural facts cut the work. The first is that the hold slot is empty only on an initial segment of the play: a move either places the arrival and leaves the hold alone, or places the held piece and puts the arrival in its place, or — only from an empty hold — stashes the arrival without placing anything. The last two leave the hold occupied, so a play stashes at most once.

Write $\mathrm{HG}_m(h; s_1 s_2 \ldots)$ for the set of length-m sequences the player can place starting from hold h with $s_1, s_2, \ldots$ arriving, each step placing either the held piece (the arrival replacing it) or the arrival (the hold unchanged).

Theorem 17. *If the arrival sequence $q_0 q_1 \ldots q_{L-1}$ has $L \geq n + 1$, then the set of placement sequences of length n the player can achieve is*

$$q_1 \cdot \mathrm{HG}_{n-1}(q_0; q_2 q_3 \ldots) \;\cup\; q_0 \cdot \mathrm{HG}_{n-1}(q_1; q_2 q_3 \ldots),$$

which is symmetric in q_0 and q_1. Exchanging the first two arrivals therefore changes neither the achievable placement sequences nor whether a perfect clear exists.

Proof. The two terms are the two continuations after stashing q_0 immediately, so they are achievable. For the converse, note that inside HG the move "place the held piece, keep the arrival" repeated from hold q_1 emits $q_1, q_2, q_3, \ldots$ in order. Since a play stashes at most once, it either never stashes, giving $q_0 q_1 \ldots q_{n-1}$, which is q_0 followed by $n - 1$ such repetitions and needs the arrival q_n — the one place where $L \geq n + 1$ is used — or stashes once, at q_k say, giving $q_0 \ldots q_{k-1}$ followed by $\mathrm{HG}_{n-k}(q_k; q_{k+1} \ldots)$; for $k = 0$ this is the displayed union, for $k = 1$ its second term, and for $k \geq 2$ it is reached inside that second term by $k - 1$ repetitions before branching freely. □

So the 5040 first bags fall into 2520 classes of two, and a bad first bag always has a bad partner. The hypothesis is not decorative: with $L = n$ no stash is possible and the play is forced, and among sequences of four pieces at width 4 there are 66 pairs differing only in their first two entries where one clears and the other does not.

The mirror of Lemma 3 acts here too: reflecting a play reflects both bags at once and permutes the 840 prefixes among themselves, so a first bag and its mirror agree prefix by prefix. Reflection commutes with the exchange, and neither fixes any first bag — the exchange because the first two pieces of a bag are distinct, the mirror because J and L each occur once — so together they cut the 5040 orders to 1260 classes of four, which is the form in which the exhaustive computation is worth attempting.

The description of plays can be sharpened, and the sharper form says how far Theorem 17 extends. Stash the first arrival at once — by the theorem this costs nothing — and call it a_0, with $a_1 a_2 \dots a_m$ arriving after. Exactly one arrived piece is pending at any time, so a play is precisely a choice of the steps $j_1 < \cdots < j_r$ at which the held piece is placed: step j_k receives $a_{j_{k-1}}$ (with $j_0 = 0$), every other step receives its own arrival, and a_{j_r} is left in the hold. One piece at a time may be deferred, and releasing it defers whatever stood in its place. In this form Theorem 17 is the bijection $C \leftrightarrow C \triangle \{1\}$ between chains.

That bijection is special to the front.

Proposition 18. *Exchanging two adjacent arrivals q_i, q_{i+1} with $i \geq 1$ does not in general preserve the achievable placement sequences, nor even the existence of a perfect clear.*

Proof. Let B be the board reached before the two steps, x the held piece and a, b the two arrivals. Two steps later the reachable states are (Ba, x) and (Bx, a) with b pending, which the exchange fixes, together with (Bab, x), which it sends to (Bba, x); and placing a before b reaches boards that placing b before a does not. At the front there is no held piece x, which is why the branches coincide there and nowhere else. For the verdict, at $w = 10$ and $n = 10$ the first bag JTLIOSZ with prefix LOZS admits no perfect clear while JLTIOSZ, its exchange of the second and third arrivals, does. □

The counterexamples are scarce: over the 250,316 decided pairs available to us the exchange preserved the verdict 250,293 times and failed 23. A rule that holds in 99.99% of a quarter of a million cases is still not a theorem. Equality of search cost is no evidence either, and fails in both directions: exchanging q_0 and q_1 agreed on the state count to the digit across 1,684 solvable cases but differed on the first failing pair (38,542,538 against 38,537,388), while the exchange of Proposition 18, which need not preserve the verdict, often preserves the count exactly — two of its four branches coincide.

One caution about the computation changed an answer for us. A first pass with a state cap of 8×10^6 reported *no* failures at all: each was sitting in a case that hit the cap, undecided. A capped search is not a negative result.

What the exhaustive computation shows so far

The sweep over the 1260 classes covers 307 of them over all 840 prefixes; for the remaining 953 we settled every class against the 14 prefixes that had produced a failure, 13,346 cases with nothing undecided. Counts below are as of that point and can only grow, since the 826 untried prefixes are untested on those 953 classes. Say that a first bag is *late* if S and Z both arrive in position five or later. Exactly 180 of the 1260 classes are late, and all of them are settled over all 840 prefixes, with nothing left undecided: of those 151,200 cases, 91 admit no perfect clear, and they fall in 31 of the 180 classes. Arranged by the position of the I piece, the failing classes are

position of I	1	3	4	5	6	7
late classes	72	36	36	12	12	12
of which fail	0	4	12	12	2	1

There is no column for position 2: by Theorem 17 the first two arrivals are interchangeable, so no class has its I there. Every late class with the I first succeeds, every late class with the I fifth fails, and the difficulty falls away again at positions six and seven. The effect is therefore not monotone in the position of the I, and the naive reading of the table — that moving the I earlier cannot hurt — is false. Moving it from sixth to fifth turns a possible position into an impossible one in 68 of the 72 matched pairs, and in none of the 72 does the earlier I help.

One family is completely decided and is the sharpest impossibility we have.

Theorem 19 (computer-verified)**.** *Let the first bag consist of T, J, L, O in any of the* 24 *orders, followed by I, S, Z, and let the second-bag prefix be a permutation of $\{J, O, S, Z\}$. Then no perfect clear exists* if and only if *the prefix ends in S and the J precedes the Z — that is, for* JOZS, JZOS *and* OJZS*, and for no other. All* 576 *cases are settled,* 288 *of them by computation and the rest by Theorem 17. Mirroring gives the same statement for the tail I, Z, S over $\{L, O, S, Z\}$: failure exactly when the prefix ends in Z and the L precedes the S, namely* LOSZ*,* LSOZ*,* OLSZ*.*

At the other end, no first bag with the I first has been made to fail: the 72 late classes of that kind are settled over all 840 prefixes, and the other 288 were tested against the six prefixes that have produced failures, all 1728 cases possible and none undecided. The most expensive settled in 7.9×10^6 states, a factor of four below the 3.2–5.6 $\times 10^7$ the known failures demand. This is a probe, not a proof: six of 840 prefixes is under one percent.

A proof of Theorem 19 by counting is not merely missing but impossible, and the obstruction disqualifies every tool in this paper.

Proposition 20. *Whether a perfect clear exists is not a function of the multiset of arriving pieces. Consequently no invariant computed from the multiset of placed pieces — column counting as in Lemma 2, the parity identity of Appendix A, or a Walkup-style argument — can decide the question.*

Proof. At $w = 10$, $n = 10$, the arrival sequences TJILSZO OJZS and TJLISZO OJZS differ by one transposition of adjacent letters and so present the same eleven pieces, hence the same candidate multisets of ten placed pieces. The first admits a perfect clear and the second does not. □

Nor does the position of the I determine the verdict on its own, once one leaves the family of Theorem 19: JTLIOSZ fails on LOZS while JLTIOSZ succeeds on it, and both have their I fourth.

Of the regularities we proposed for the failing prefixes, one survives: across all 108 failures known to us the prefix is a permutation of $\{O, S, Z\}$ together with one of J or L, without exception, which is the shape the mirror lemma forces. Three others broke, each having held for every case then available — that the prefixes end in ZS, that they end in S or Z (the endings now stand at S sixty times, Z forty-three and O five), and that they take only three shapes.

A fourth broke on the 953 classes probed last. We had spoken of "the prefixes that produce failures" as though the set were a property of the prefix alone. It is not: the failing prefixes depend on the first bag. The class STIZLOJ fails on OJSZ and JOSZ and succeeds on the other twelve prefixes tried, among them OJZS, JOZS and JZOS, the three that Theorem 19 kills for every first bag in its family. Nor is the split stable under transposition: STLZJIO fails on JOSZ while STLZIJO, which differs by exchanging the fourth and fifth arrivals, does not. Of the 120 classes beginning S, T we settled 119 against all 14 prefixes; 12 cases fail, in 9 classes, and every

failing prefix is OJSZ or JOSZ. The remaining 111 classes admit a perfect clear on all 14, so these classes form no family of their own.

The rule of Theorem 19 does not survive a change of alphabet either. Over $\{L, O, S, Z\}$, with the tail I, S, Z unchanged — a region neither the theorem nor its mirror image covers — the same shape of condition bounds the failures without determining them: TJLOISZ and OTJLISZ both fail on LOZS and OLZS, which end in S with the L before the Z, but only the second fails on LZOS, which satisfies the condition as well.

The most instructive was the lateness condition. Every failing first bag then known was late; of a hundred sampled bags, fifteen were late and only three of those failed; and a prediction recorded before the last undecided cases were settled — that they would succeed, none being late — came out right. It survived until the 180 late classes were done and the sweep moved on, where JTZOLSI OJZS and OSTLJZI OJSZ fail with Z third and S second. The prediction had in fact tested something else: all four of those cases had the I first, which the table above shows to be the easy end. A confirmed prediction need not confirm the reason it was made for.

The mechanism one would guess for lateness remains natural — a vertical I is the main way to resolve an S/Z staircase inside four rows, and an I arriving after the staircase is built cannot undo it, the direction Burgiel's theorem [6] points in; see also [7]. The computation says the guess does not survive contact with the position of the I.

We claim no novelty for this section. The community's own perfect-clear solvers work in a more general model — ours is hard-drop only, so an impossibility here is an impossibility *in our model*, and spins or tucks might rescue some of these cases. The point is the shape of the answer: perfect clears are almost always available, failures are rare and structured, and the well-known openers exist to save search and memorisation rather than to reach positions that are otherwise unreachable.

A A parity identity

The following is folklore among competitive players, where it is used to decide whether a perfect clear is still reachable; see the Hard Drop wiki [12]. We record it because it has a short proof, generalises to every width, and explains a natural false conjecture.

Colour the board like a chessboard and set

$$\varphi(B) = \sum_{(r,c)\in B} (-1)^{r+c}.$$

Every tetromino except T covers two cells of each colour, so placing it leaves φ unchanged; a T covers three of one colour and one of the other, changing φ by ± 2, hence by 2 mod 4.

Consider a deletion event removing full rows $j_1 < \cdots < j_k$. For a surviving cell at row r, let $s(r)$ be the number of deleted rows below it; the cell descends by $s(r)$, multiplying its contribution by $(-1)^{s(r)}$. Put

$$D = \#\{\text{surviving cells with } s(r) \text{ odd}\}, \qquad A = \sum_{i=1}^{k} \varphi(\text{row } j_i),$$

noting that φ of a full row is 0 when w is even and $(-1)^j$ when w is odd. Then

$$\Delta\varphi = -2 \sum_{s(r)\ \text{odd}} (-1)^{r+c} \;-\; A \equiv -2D - A \pmod 4,$$

since modulo 4 the first sum depends only on the parity of the number of cells involved. Summing over a whole perfect clear, where φ starts and ends at 0, gives $2\,\#\mathrm{T} \equiv 2\sum D + \sum A \pmod 4$, that is

$$\#\mathrm{T} \;\equiv\; \sum_{\text{events}} D \;+\; \tfrac{1}{2} \sum_{\text{events}} A \pmod 2.$$

For even w the correction A vanishes and the identity reads: *the parity of the number of T pieces equals the parity of the total number of cells left above the deleted rows.* In particular a perfect clear with no deletions at all uses an even number of T pieces — which is the false conjecture one arrives at by forgetting that rows disappear.

Remark 21. D must be counted per deleted row, or equivalently as the number of cells with an odd number of deleted rows below them. Reading it as "the cells above the deleted rows" is correct only for single-row deletions.

Computations

All constructions and all counting claims in this paper are reproducible from the accompanying repository, which contains for each numerical statement the command that produces it. The identity of Appendix A was checked against all 4,758,805 perfect clears at widths 4, 5, 6, 7 and 8, by two independent implementations agreeing on the number of plays in every case.

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